

Is SPAR-PIBT better? One cell per rival per map

Each cell is SPAR-PIBT's throughput divided by that rival's on that map: above 1.00x SPAR delivers more, below 1.00x it delivers less. Against both published baselines SPAR wins everywhere by 13x to 315x. Against the plain PIBT it extends it is ahead on the two aisle-constrained maps and behind on the two open ones -- and at five seeds only the two losses clear $p < 0.05$.

	plain lifelong PIBT	Token Passing	Token Passing + recovery	RHCR
bottleneck aisle-constrained	1.27x ahead, not significant 149 vs 118 per 1000 steps $p = 0.056$	20x SPAR ahead 149 vs 8 per 1000 steps $p = 0.008$	18x SPAR ahead 149 vs 8 per 1000 steps $p = 0.008$	13x SPAR ahead 149 vs 12 per 1000 steps $p = 0.008$
corridors aisle-constrained	1.21x ahead, not significant 158 vs 130 per 1000 steps $p = 0.063$	delivers nothing at all SPAR wins 158 vs 0 per 1000 steps $p = 0.008$	delivers nothing at all SPAR wins 158 vs 0 per 1000 steps $p = 0.008$	315x SPAR ahead 158 vs 0 per 1000 steps $p = 0.008$
narrow open floor	0.82x SPAR behind 289 vs 354 per 1000 steps $p = 0.016$	not run on this map	not run on this map	not run on this map
medium open floor	0.81x SPAR behind 405 vs 502 per 1000 steps $p = 0.008$	101x SPAR ahead 405 vs 4 per 1000 steps $p = 0.008$	15x SPAR ahead 405 vs 27 per 1000 steps $p = 0.008$	28x SPAR ahead 405 vs 14 per 1000 steps $p = 0.008$